This guide provides step-by-step solutions for your **Business Finance Assignment on the Time Value of Money**, using standard financial formulas for compounding and discounting.

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1. Future Value of a Single Deposit

Question: If you deposit $2,000 in a bank account that pays 6% interest annually, how much will be in your account after 5 years?

- **Formula:** $FV = PV \times (1 + i)^n$
- **Variables:**
  - PV (Present Value) = $2,000

  - i (Interest Rate) = 0.06 (6%)

  - n (Number of Years) = 5

- **Calculation:**
  $FV = 2,000 \times (1 + 0.06)^5$
  $FV = 2,000 \times 1.3382$
  $FV = \$2,676.45$

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2. Present Value of a Future Payment

Question: What is the present value of a security that will pay $29,000 in 20 years if securities of equal risk pay 5% annually?

- **Formula:** $PV = \frac{FV}{(1 + i)^n}$
- **Variables:**
  - FV (Future Value) = $29,000

  - i (Interest Rate) = 0.05 (5%)

  - n (Number of Years) = 20

- **Calculation:**
  $PV = \frac{29,000}{(1 + 0.05)^{20}}$
  $PV = \frac{29,000}{2.6533}$
  $PV = \$10,929.79$

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3. Solving for Time (Rule of 72)

Question: If you deposit money today in an account that pays 4% annual interest, how long will it take to double your money?

- **Formula:** $n = \frac{\ln(FV/PV)}{\ln(1 + i)}$
- **Variables:**
  - Target: Double the money (If $PV=1$, then $FV=2$)
  - i (Interest Rate) = 0.04 (4%)
- **Calculation:**
  $n = \frac{\ln(2)}{\ln(1.04)} \approx \frac{0.6931}{0.0392}$
  $n \approx 17.67 \text{ years}$
  (Quick estimate using the Rule of 72: $72 / 4 = 18 \text{ years}$)

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4. Solving for Interest Rate

Question: Your parents have $350,000 and need $800,000 in 19 years. What annual interest rate must they earn?

- **Formula:** $i = \left(\frac{FV}{PV}\right)^{1/n} - 1$
- **Variables:**
  - PV = $350,000
  - FV = $800,000
  - n = 19
- **Calculation:**
  $i = (800,000 / 350,000)^{1/19} - 1$
  $i = (2.2857)^{0.0526} - 1$
  $i = 1.0445 - 1$
  $i = 4.45\%$

5. Comparative Compounding and Discounting

Find the following values (Compounded annually):

- a. $200 compounded for 10 years at 4%: $FV = 200 \times (1.04)^{10} = 200 \times 1.4802 = \mathbf{\$296.05}$
- b. $200 compounded for 10 years at 8%: $FV = 200 \times (1.08)^{10} = 200 \times 2.1589 = \mathbf{\$431.78}$
- c. Present Value of $200 due in 10 years at 4%: $PV = 200 / (1.04)^{10} = 200 / 1.4802 = \mathbf{\$135.11}$
- d. Present Value of $1,870 due in 10 years:
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  - **At 8%:** $PV = 1,870 / (1.08)^{10} = 1,870 / 2.1589 = \mathbf{\$866.18}$
  - **At 4%:** $PV = 1,870 / (1.04)^{10} = 1,870 / 1.4802 = \mathbf{\$1,263.34}$
- e. Definitions and Effects:
  - **Definition:** Present Value is the current value of a future sum of money or stream of cash flows given a specific rate of return.
  - **Interest Rate Effect:** There is an **inverse relationship** between interest rates and Present Value. As the interest rate increases, the Present Value decreases (as seen in part d where 8% yielded a lower PV than 4%).
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6. Basic Equations (Short-Term)

Find the following values using compounding/discounting equations:

- a. $600 compounded for 1 year at 6%: $FV = 600 \times (1.06)^1 = \mathbf{\$636.00}$
- b. $600 compounded for 2 years at 6%: $FV = 600 \times (1.06)^2 = \mathbf{\$674.16}$

- c. PV of \$600 due in 1 year at 6%: $PV = 600 / (1.06)^1 = \mathbf{\$566.04}$
- d. PV of \$600 due in 2 years at 6%: $PV = 600 / (1.06)^2 = \mathbf{\$534.00}$